

Soroush Mehraban

Toronto, ON | soroush.mehraban@mail.utoronto.ca | soroushmehraban.github.io
LinkedIn | GitHub | Google Scholar

SUMMARY

Generative AI and 3D computer vision researcher completing a PhD at the University of Toronto and working as a Research Scientist Intern at Pickford AI. Builds diffusion and transformer systems for text-to-3D scene generation, video style transfer, co-speech gesture synthesis, human mesh recovery, and motion understanding. Author of 11 publications at venues including NeurIPS, ICCV, WACV, ICPR, FG, and ECCV workshops.

EXPERIENCE

Pickford AI

Research Scientist Intern

May 2025 – Present

Remote / Los Angeles, CA

- Developing generative models for text-to-3D scene generation.
- Built **PickStyle**, a video-to-video diffusion framework using low-rank context-style adapters, motion-augmented image pairs, and factorized classifier-free guidance; accepted to the ECCV 2026 AI4VA Workshop.
- Co-developed **Puppeteer**, a posture-aware, object-grounded co-speech gesture model conditioned on audio, text, motion history, body posture, and surrounding 3D geometry.

University of Toronto

PhD Researcher

Sep 2022 – Present

Toronto, ON

- Developed **FastHMR**, combining error-constrained layer merging, mask-guided token merging, and diffusion decoding to deliver up to **2.3× faster inference** with slightly improved accuracy; WACV 2026.
- Co-created **3DPain**, a balanced synthetic dataset with 82,500 frames across 2,500 identities, and ViTPain, a reference-guided transformer for automated pain assessment; ICPR 2026.
- Faculty Affiliate Researcher at the Vector Institute; published work spanning generative models, 3D pose and mesh recovery, action recognition, implicit representations, and clinical motion analysis.

SELECTED FIRST-AUTHOR PUBLICATIONS

- **PickStyle: Video-to-Video Style Transfer with Context-Style Adapters** ECCV 2026 AI4VA Workshop
- **FastHMR: Accelerating Human Mesh Recovery via Token and Layer Merging** WACV 2026
- **STARS: Self-Supervised 3D Action Recognition with Contrastive Tuning** WACV 2026
- **MotionAGFormer: Transformer-GCNFormer for 3D Human Pose Estimation** WACV 2024

EDUCATION

University of Toronto

PhD, Biomedical Engineering

Expected Sep 2026

Toronto, ON

Amirkabir University of Technology

BSc, Computer Engineering | GPA: 19.15/20 (4.00/4.00)

Jul 2022

Tehran, Iran

TECHNICAL SKILLS

ML/CV: PyTorch, OpenCV, scikit-learn, diffusion models, transformers, video generation, 3D vision, self-supervised learning

Programming: Python, C, Java, C# | **Data:** NumPy, pandas, Matplotlib | **Tools:** Docker, Git, Linux

LEADERSHIP & COMMUNICATION

Teaching Assistant, University of Toronto

2024 – 2025

Delivered computer vision lectures and tutorials and developed course material on transformers, video tracking, activity recognition, and human body tracking.